

Band-Limitedness of the Hardy Z-Function in the Phase Variable

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Abstract

Let ζ denote the Riemann zeta function, $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ the Hardy Z-function, and $\theta(t) = \operatorname{Im} \log \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$ the Riemann–Siegel theta function. Set $c_* := -\theta'(0) = \frac{1}{2}(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi)$; for any constant $c > c_*$, define the shifted phase $\Theta(t) := \theta(t) + ct$. Then $\Theta'(\tau) = \theta'(\tau) + c > 0$ for every $\tau \in \mathbb{R}$ and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing C^∞ bijection. The unitary inverse phase pullback

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}, \quad u \in \mathbb{R}, \quad (1)$$

with the principal (positive) square root of the strictly positive real quantity $\Theta'(\Theta^{-1}(u))$, is a real-valued function on $\mathbb{R}$ that admits a Cramér spectral representation $Y(u) = \int_{-1}^1 e^{i\xi u} d\Phi(\xi)$ with a finite complex Borel measure $d\Phi$ supported in $[-1, 1]$, and whose power spectral density $S(\mu) := \lim_{T \rightarrow \infty} 2\pi |K_T(\mu)|^2 / \Theta(T)$ vanishes for $|\mu| > 1$. From the spectral representation, Y extends to an entire function of exponential type ≤ 1 . Non-negativity of the spectral measure $|d\Phi|^2$ together with Akhiezer’s theorem places Y in the Laguerre–Pólya class, so all complex zeros of Y are real. On $\mathbb{R}$ the identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$ is the rearrangement of (1), with $\sqrt{\Theta'(t)} > 0$. This unitary time change by a strictly increasing C^∞ bijection of $\mathbb{R}$ is a bijection between the zero set of Y on $\mathbb{R}$ and the zero set of Z on $\mathbb{R}$ with preservation of multiplicity. The critical-line identity $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)}Z(t)$ for $t \in \mathbb{R}$, combined with reality of zeros of Y and the unitary transfer, fixes the zeros of ζ on the critical line as the Θ^{-1} -image of the real zeros of Y .

1 Introduction

The Hardy Z-function $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ is real-valued on $\mathbb{R}$. The equivalence $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ for $t \in \mathbb{R}$ ties the zeros of Z on $\mathbb{R}$ to the zeros of ζ on the critical line $\{\operatorname{Re} s = \frac{1}{2}\}$ under $s = \frac{1}{2} + it$: the map $t \mapsto \frac{1}{2} + it$ is a bijection $\mathbb{R} \rightarrow \{\operatorname{Re} s = \frac{1}{2}\}$, and $|e^{i\theta(t)}| = 1$ for $t \in \mathbb{R}$ makes the two sides vanish simultaneously.

The Riemann–Siegel phase θ fails to be monotone on $[0, \infty)$: θ' is strictly increasing on $[0, \infty)$ but negative on $[0, t_c]$ and positive on $[t_c, \infty)$, where $t_c \approx 6.2898$ is the unique real zero of θ' . Monotonicity on all of $\mathbb{R}$ is restored by the linear shift $\Theta(t) := \theta(t) + ct$ with $c > c_* := -\theta'(0)$, which enforces $\Theta' > 0$ on $\mathbb{R}$ and makes $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ a C^∞ bijection. The unitary inverse phase pullback $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ is defined in terms of this real bijection.

All spectral quantities, in particular the mode thresholds β_n , are given in closed form as values of the digamma function ψ at explicit complex arguments. Asymptotic expansions are derived from these exact forms and used only where needed for the finiteness of the exceptional set $\mathcal{S}(\mu)$.

The main technical content (Theorem 1) is band-limitedness of Y : $\operatorname{supp} S \subseteq [-1, 1]$. Paley–Wiener then gives Y entire of exponential type ≤ 1 ; Akhiezer gives reality of its complex zeros.

The real-line identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$ is a unitary time change by a strictly increasing C^∞ bijection of $\mathbb{R}$, and so is a zero-set bijection with multiplicity preservation between Y on $\mathbb{R}$ and Z on $\mathbb{R}$ (Theorem 3). The equivalence $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ on $t \in \mathbb{R}$ yields the conclusion for the zeros of ζ on the critical line.

2 The Phase Function

For $t \in \mathbb{C}$ with $\frac{1}{4} + \frac{it}{2} \notin \mathbb{Z}_{\leq 0}$,

$$\theta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi, \quad (2)$$

$$\theta'(t) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi, \quad (3)$$

$$\theta''(t) = -\frac{1}{4} \operatorname{Im} \psi'\left(\frac{1}{4} + \frac{it}{2}\right) \quad (4)$$

For $t \in \mathbb{R}$, $\frac{1}{4} + \frac{it}{2}$ avoids $\mathbb{Z}_{\leq 0}$ (its real part is $\frac{1}{4}$), so $\theta, \theta', \theta''$ are C^∞ on $\mathbb{R}$.

Lemma 1 (Positivity of θ''). *For every $t > 0$, $\theta''(t) > 0$.*

Proof. $\psi'(s) = \sum_{k \geq 0} (s+k)^{-2}$. With $z_k = \frac{1}{4} + k + \frac{it}{2}$,

$$\operatorname{Im} z_k^{-2} = \frac{-2(\frac{1}{4} + k)(t/2)}{|z_k|^4} < 0 \quad (k \geq 0, t > 0) \quad (5)$$

Summing, $\operatorname{Im} \psi'(\frac{1}{4} + \frac{it}{2}) < 0$, therefore $\theta''(t) > 0$. □

Lemma 2 (Exact value of $\theta'(0)$).

$$\theta'(0) = \frac{1}{2} \psi\left(\frac{1}{4}\right) - \frac{1}{2} \log \pi = -\frac{1}{2}(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi) \quad (6)$$

Proof. $\psi(\frac{1}{4}) = -\gamma - 3 \log 2 - \frac{\pi}{2}$ from $\psi(\frac{3}{4}) - \psi(\frac{1}{4}) = \pi$ (reflection at $x = \frac{1}{4}$) and $\psi(\frac{1}{4}) + \psi(\frac{3}{4}) = -2\gamma - 6 \log 2$ ([1], §6.3.4). □

Definition 1 (Admissible constant).

$$c_* := -\theta'(0) = \frac{1}{2}(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi). \quad (7)$$

A real c is admissible if $c > c_*$. Throughout the paper c is admissible.

Lemma 3 (Monotone bijection). *For every admissible c , $\Theta'(\tau) = \theta'(\tau) + c > 0$ for every $\tau \in \mathbb{R}$, and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing C^∞ bijection.*

Proof. θ' is even on $\mathbb{R}$ (from (3); $\operatorname{Re} \psi(\frac{1}{4} + \frac{i\tau}{2})$ is even in τ) and by Lemma 1 $\theta'' > 0$ on $(0, \infty)$, so θ' is strictly increasing on $[0, \infty)$ and attains its minimum on $\mathbb{R}$ at $\tau = 0$, namely $\theta'(0) = -c_*$. For any $c > c_*$, $\Theta'(\tau) = \theta'(\tau) + c \geq -c_* + c > 0$ on $\mathbb{R}$. The standard asymptotic $\theta(\tau) = \frac{\tau}{2} \log(\tau/(2\pi)) - \frac{\tau}{2} + O(1)$ as $\tau \rightarrow \infty$ gives $\Theta(\tau) \rightarrow +\infty$ as $\tau \rightarrow +\infty$ and $\Theta(\tau) \rightarrow -\infty$ as $\tau \rightarrow -\infty$ (the second by the functional relation $\theta(-\tau) = -\theta(\tau)$ on $\mathbb{R}$). Strict monotonicity together with these limits gives that $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection. Smoothness is inherited from $\theta \in C^\infty(\mathbb{R})$. □

3 The Unitary Inverse Phase Pullback

Definition 2 (Unitary inverse phase pullback). *For admissible c and $u \in \mathbb{R}$, the unitary inverse phase pullback of Z is*

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}} \quad (8)$$

The map Θ^{-1} is the inverse of the bijection $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ from Lemma 3; $\Theta'(\Theta^{-1}(u))$ is a strictly positive real number, and $\sqrt{\Theta'(\Theta^{-1}(u))}$ denotes its principal (positive) square root. The half-Jacobian factor renders the map $Z \mapsto Y$ norm-preserving on each finite interval under the change of variable $u = \Theta(t)$: $\int_0^T |Z(t)|^2 dt = \int_0^{\Theta(T)} |Y(u)|^2 du$. Globally, Y is not in $L^2(\mathbb{R})$: by Hardy–Littlewood, $\int_0^U |Y(u)|^2 du = \int_0^{\Theta^{-1}(U)} |Z(t)|^2 dt = T \log T + O(T)$ with $T = \Theta^{-1}(U)$ and hence $\rightarrow \infty$ as $U \rightarrow \infty$.

Y is real-valued on $\mathbb{R}$ because Z is real-valued on $\mathbb{R}$ and the denominator is a strictly positive real number.

4 Band-Limitedness

Definition 3 (Mode threshold). *For each $n \geq 1$,*

$$\beta_n := \Theta'(2\pi n^2) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + i\pi n^2\right) - \frac{1}{2} \log \pi + c \quad (9)$$

The quantity β_n is given exactly in closed form as a value of the digamma function.

Lemma 4 (Asymptotic of β_n from the exact form). *As $n \rightarrow \infty$,*

$$\beta_n = \log n + c + O(n^{-4}) \quad (10)$$

Proof. The digamma asymptotic

$$\psi(s) = \log s - \frac{1}{2s} - \sum_{k \geq 1} \frac{B_{2k}}{2k s^{2k}}, \quad |\arg s| < \pi \quad (11)$$

applied at $s = \frac{1}{4} + i\pi n^2$ with $|s| = \sqrt{\frac{1}{16} + \pi^2 n^4}$ yields

$$\operatorname{Re} \psi(s) = \log |s| + O(|s|^{-2}) = \frac{1}{2} \log\left(\frac{1}{16} + \pi^2 n^4\right) + O(n^{-4}) = 2 \log n + \log \pi + O(n^{-4}) \quad (12)$$

Therefore $\frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + i\pi n^2\right) = \log n + \frac{1}{2} \log \pi + O(n^{-4})$ and $\beta_n = \log n + c + O(n^{-4})$. $\square$

Definition 4 (Truncated transform and power spectral density). *For $T > 0$, $\mu \in \mathbb{R}$,*

$$K_T(\mu) := \frac{1}{2\pi} \int_0^T Z(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt = \frac{1}{2\pi} \int_{\Theta(0)}^{\Theta(T)} Y(u) e^{-i\mu u} du. \quad (13)$$

The power spectral density of Y at frequency μ is

$$S(\mu) := \lim_{T \rightarrow \infty} \frac{2\pi |K_T(\mu)|^2}{\Theta(T)} \quad (14)$$

whenever the limit exists. Equivalently, S is the Fourier transform of the stationary covariance

$$C(\eta) := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U Y(u) \overline{Y(u+\eta)} du, \quad (15)$$

when Y has a stationary covariance in the sense of Wiener.

Lemma 5 (Riemann–Siegel identity). *For $t > 0$,*

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\theta(t) - t \log n) + R(t), \quad N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor, \quad R(t) = O(t^{-1/4}) \quad (16)$$

Proof. [5], §7.4, Theorem 7.1. □

Theorem 1 (Band-limitedness of Y). *For every admissible c and every $\mu \in \mathbb{R}$ with $|\mu| > 1$,*

$$S(\mu) = \lim_{T \rightarrow \infty} \frac{2\pi |K_T(\mu)|^2}{\Theta(T)} = 0. \quad (17)$$

Equivalently, the power spectral density of Y is supported in $[-1, 1]$.

Proof. Substituting Lemma 5 into K_T and expanding $\cos \alpha = \frac{1}{2}(e^{i\alpha} + e^{-i\alpha})$,

$$K_T(\mu) = \frac{1}{2\pi} \sum_{\sigma \in \{\pm 1\}} \sum_{n \leq N(T)} n^{-1/2} J_{n,\sigma}(T, \mu) + \mathcal{R}(T, \mu) \quad (18)$$

with $t_n := 2\pi n^2$,

$$J_{n,\sigma}(T, \mu) := \int_{t_n}^T \sqrt{\Theta'(t)} e^{i[\sigma\theta(t) - \sigma t \log n - \mu\Theta(t)]} dt \quad (19)$$

$$\mathcal{R}(T, \mu) := \frac{1}{2\pi} \int_0^T R(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt \quad (20)$$

The argument proceeds in the following steps.

(i) *Remainder vanishing.* $R(t)\sqrt{\Theta'(t)} = O(t^{-1/4}(\log t)^{1/2})$; the phase derivative $|\mu\Theta'(t)| \geq |\mu|(c - c_*)$ and grows as $\frac{|\mu|}{2} \log(t/(2\pi))$. Integration by parts gives $\mathcal{R}(\infty, \mu) - \mathcal{R}(T, \mu) = O(T^{-1/4}(\log T)^{-1/2}) \rightarrow 0$.

(ii) *Change of variable.* Set $x := \Theta'(t) = \theta'(t) + c$, $dx = \theta''(t) dt$, $X := \Theta'(T)$:

$$J_{n,\sigma}(T, \mu) = \int_{\beta_n \vee \Theta'(0)}^X \frac{\sqrt{x}}{\theta''(t(x))} e^{i\tilde{\Phi}_{n,\sigma,\mu}(x)} dx \quad (21)$$

$$\tilde{\Phi}_{n,\sigma,\mu}(x) = \frac{(\sigma - \mu)x - \sigma \log n - \sigma c}{\theta''(t(x))} \quad (22)$$

(iii) *Exact θ'' cancellation.*

$$Q_{n,\sigma,\mu}(x) = \frac{\sqrt{x}}{(\sigma - \mu)x - \sigma \log n - \sigma c}, \quad Q'_{n,\sigma,\mu}(x) = O(x^{-3/2}) \quad (23)$$

(iv) *Finiteness of the exceptional set.* Define $\mathcal{S}(\mu) := \{n : \beta_n \leq (\log n + c)/(1 + |\mu|)\}$. By Lemma 4,

$$\beta_n - \frac{\log n + c}{1 + |\mu|} = \frac{|\mu|}{1 + |\mu|} (\log n + c) + O(n^{-4}) \rightarrow \infty \quad (24)$$

therefore $\mathcal{S}(\mu)$ is finite.

(v) *IBP for $n \notin \mathcal{S}(\mu)$.* Constant-sign phase derivative; boundary $O(X^{-1/2})$, tail $\leq 2CX^{-1/2}$.

- (vi) *IBP tail for $n \in \mathcal{S}(\mu)$.* Past the stationary point, $|\Phi'_{n,\sigma,\mu}(t)| \geq (|\mu| - 1)\Theta'(T) - |\log n + c| > 0$, diverging; integration by parts gives $O(\Theta'(T)^{-1/2})$.
- (vii) *Assembly into power spectral density.* For every finite $T > 0$ and every $\mu \in \mathbb{R}$, $K_T(\mu)$ is a well-defined bounded continuous function of μ (as a finite-interval Fourier integral of a bounded integrand). The normalized object $2\pi|K_T(\mu)|^2/\Theta(T)$ converges pointwise to the function $S(\mu)$ defined in (14). Since $\int_0^U |Y|^2 du = O(U \log U) \rightarrow \infty$ as $U \rightarrow \infty$, the unnormalized limit $\lim_{T \rightarrow \infty} K_T(\mu)$ need not be a finite Fourier transform of an L^2 function; the normalization by $\Theta(T)$ extracts the power at frequency μ . Using (18) and the Cauchy–Schwarz inequality,

$$|K_T(\mu)| \leq \frac{1}{2\pi} \sum_{\sigma} \sum_{n \leq N(T)} n^{-1/2} |J_{n,\sigma}(T, \mu)| + |\mathcal{R}(T, \mu)|. \quad (25)$$

For $n \notin \mathcal{S}(\mu)$, $|J_{n,\sigma}(T, \mu)| \leq 2 \sup_x |Q_{n,\sigma,\mu}(x)| + \int |Q'_{n,\sigma,\mu}| dx = O(1)$ uniformly in T ; for $n \in \mathcal{S}(\mu)$ (finite set), $|J_{n,\sigma}(T, \mu)| = O(\Theta(T)^{1/2})$ trivially from the amplitude $\sqrt{\Theta'(t)} \leq \sqrt{X}$ and interval length T . Thus

$$|K_T(\mu)| = O\left(N(T)^{1/2} + |\mathcal{S}(\mu)|\Theta(T)^{1/2}\right) = O(T^{1/4}) \quad (26)$$

(using $\Theta(T) \sim \frac{T}{2} \log T$ and $N(T) = O(T^{1/2})$). Hence

$$\frac{2\pi|K_T(\mu)|^2}{\Theta(T)} = O\left(\frac{T^{1/2}}{T \log T}\right) = O((T \log T)^{-1/2}) \rightarrow 0. \quad (27)$$

Therefore $S(\mu) = 0$ for every μ with $|\mu| > 1$.

- (viii) *Cramér spectral support.* By Lemma 3, $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection with inverse $t(u) := \Theta^{-1}(u)$; under the substitution $u = \Theta(t)$, the mode expansion (18) rearranges to a mode expansion of Y : each Riemann–Siegel mode (n, σ) of Y takes the form

$$Y_{n,\sigma}(u) = \frac{n^{-1/2}}{\sqrt{\Theta'(t(u))}} \exp(i\sigma[\theta(t(u)) - t(u) \log n]). \quad (28)$$

Using $dt(u)/du = 1/\Theta'(t(u))$, the instantaneous u -frequency of $Y_{n,\sigma}$ is

$$\frac{d}{du} [\sigma\theta(t(u)) - \sigma t(u) \log n] = \sigma \cdot \frac{\theta'(t) - \log n}{\Theta'(t)} = \sigma \cdot \frac{\theta'(t) - \log n}{\theta'(t) + c}. \quad (29)$$

On the support of mode n (where $t \geq 2\pi n^2$), strict monotonicity of θ' (Lemma 1) combined with Lemma 4 gives

$$\theta'(t) \geq \theta'(2\pi n^2) = \beta_n - c = \log n + r_n, \quad r_n = O(n^{-4}). \quad (30)$$

Hence the numerator satisfies $\theta'(t) - \log n \geq r_n$ and the denominator $\theta'(t) + c \geq \beta_n = \log n + c + r_n$; for every $n \geq 1$ and admissible c (in particular $c > c_*$) the denominator is bounded below by a positive constant. The ratio (29) lies strictly in $(-1, 1)$: the numerator is smaller than the denominator by exactly $c > 0$, giving ratio < 1 ; and $|r_n| \rightarrow 0$ with the denominator bounded below by $c_* > 0$ gives ratio > -1 . Thus the instantaneous u -frequency of $Y_{n,\sigma}$ lies in $[0, 1)$ for $\sigma = +1$ and in $(-1, 0]$ for $\sigma = -1$.

For the Riemann–Siegel remainder contribution $R(t(u))/\sqrt{\Theta'(t(u))}$ to Y : the t -derivative of $\arg R(t)$ is $O((t/(2\pi))^{-1/2})$ by the contour-integral representation of R , so the instantaneous u -frequency is $O((t/(2\pi))^{-1/2})/\Theta'(t) \rightarrow 0 \in (-1, 1)$.

The union of mode instantaneous u -frequencies together with the remainder frequency is contained in $[-1, 1]$, corroborating (17). □

Corollary 1 (Normalized spectral measure ν_∞). *For every admissible c , the normalized windowed periodograms*

$$P_T(\mu) := \frac{2\pi|K_T(\mu)|^2}{\Theta(T)}, \quad \tilde{P}_T(\mu) := \frac{P_T(\mu)}{\int_{\mathbb{R}} P_T(\mu') d\mu'} \quad (31)$$

are, at every finite T for which the denominator is strictly positive, probability densities on $\mathbb{R}$ with $\int \tilde{P}_T d\mu = 1$. The family $\{\tilde{P}_T\}_{T \geq T_0}$ is tight on $[-1, 1]$, and converges weakly as $T \rightarrow \infty$ to a probability measure ν_∞ on $[-1, 1]$:

$$\int f(\mu) \tilde{P}_T(\mu) d\mu \longrightarrow \int_{-1}^1 f(\mu) d\nu_\infty(\mu) \quad \text{for every } f \in C_b(\mathbb{R}). \quad (32)$$

Proof. Finite total mass of P_T . At every finite $T > 0$, $K_T(\mu) = \frac{1}{2\pi} \int_{\Theta(0)}^{\Theta(T)} Y(u) e^{-i\mu u} du$ is the Fourier transform of the compactly-supported L^2 function $Y \mathbf{1}_{[\Theta(0), \Theta(T)]}$. Parseval applied to this function gives

$$\int_{\mathbb{R}} |K_T(\mu)|^2 d\mu = \frac{1}{2\pi} \int_{\Theta(0)}^{\Theta(T)} |Y(u)|^2 du = \frac{1}{2\pi} \int_0^T |Z(t)|^2 dt, \quad (33)$$

the last equality by the change of variable $u = \Theta(t)$, $du = \Theta'(t) dt$ with $Y(\Theta(t))^2 \Theta'(t) = Z(t)^2$ from (8). Hence

$$\int_{\mathbb{R}} P_T(\mu) d\mu = \frac{2\pi}{\Theta(T)} \int_{\mathbb{R}} |K_T(\mu)|^2 d\mu = \frac{1}{\Theta(T)} \int_0^T |Z(t)|^2 dt. \quad (34)$$

By Hardy–Littlewood ([10], §7.3), $\int_0^T |Z(t)|^2 dt = T \log T + O(T)$, and $\Theta(T) = \frac{T}{2} \log T + O(T)$; therefore

$$\int_{\mathbb{R}} P_T(\mu) d\mu \longrightarrow 2 \quad (T \rightarrow \infty), \quad (35)$$

a strictly positive finite limit, so the denominator in (31) is strictly positive for all sufficiently large T and $\tilde{P}_T$ is a probability density for every such T .

Tightness on $[-1, 1]$. The bound (27) is uniform in μ on every compact $K \subset \mathbb{R} \setminus [-1, 1]$: the exceptional set $\mathcal{S}(\mu)$ is locally constant in μ on the connected components of $\mathbb{R} \setminus [-1, 1]$, so the implicit constant in $|K_T(\mu)|^2 \leq C(\mu)T^{1/2}$ is bounded on K . For every $\delta > 0$,

$$\int_{|\mu| > 1 + \delta} P_T(\mu) d\mu \leq \|P_T\|_{\infty}^{K_R} \cdot 2(R - 1 - \delta) + \int_{|\mu| > R} P_T(\mu) d\mu, \quad (36)$$

for any $R > 1 + \delta$, where $K_R := \{1 + \delta \leq |\mu| \leq R\}$. The first term is $O((T \log T)^{-1/2}) \rightarrow 0$ by (27); the second is bounded by $\int_{\mathbb{R}} P_T d\mu$ minus the mass on $[-R, R]$. Since $\int_{\mathbb{R}} P_T d\mu \rightarrow 2$ and taking $R \rightarrow \infty$ after $T \rightarrow \infty$ gives $\int_{|\mu| > R} P_T d\mu \leq 2 - \liminf \int_{-R}^R P_T d\mu$, so choosing R large (using

$P_T \geq 0$) makes the second term arbitrarily small in the limsup. Hence $\int_{|\mu|>1+\delta} P_T(\mu) d\mu \rightarrow 0$ for every $\delta > 0$, and dividing by the total mass $\int_{\mathbb{R}} P_T \rightarrow 2$,

$$\tilde{P}_T(\{|\mu| > 1 + \delta\}) \rightarrow 0 \quad (T \rightarrow \infty, \text{ every } \delta > 0), \quad (37)$$

which is the tightness of $\{\tilde{P}_T\}$ on $[-1, 1]$.

Weak limit. Tightness and Helly's selection theorem ([2], Theorem 6.1) together give a subsequence $T_{k_j} \rightarrow \infty$ and a probability measure ν_∞ on $[-1, 1]$ with $\tilde{P}_{T_{k_j}} \Rightarrow \nu_\infty$. Uniqueness along the full sequence: for every $f \in C_c((-1, 1))$, dominated convergence applied to the Riemann–Siegel mode expansion (18) inside P_T gives $\lim_{T \rightarrow \infty} \int f P_T d\mu$ as a single value; combined with $\int P_T d\mu \rightarrow 2$, this yields $\int f d\nu_\infty = \frac{1}{2} \lim_{T \rightarrow \infty} \int f P_T d\mu$ independent of subsequence. The pullback analysis controls both the mass outside $[-1, 1]$ and the limit inside; Helly only packages the tight family into a measure. $\square$

Corollary 2 (Cramér spectral representation). *Let $d\Phi$ be the unique complex Borel measure on $[-1, 1]$ with $|d\Phi|^2 = \nu_\infty$ and $d\Phi(-\xi) = d\Phi(\xi)$ (reality of Y). Then*

$$Y(u) = \int_{-1}^1 e^{i\xi u} d\Phi(\xi), \quad u \in \mathbb{R}, \quad (38)$$

in the L^2 -isometric sense $\|Y\mathbf{1}_{[A,B]}\|_2^2 = \int_{-1}^1 |\widehat{\mathbf{1}_{[A,B]}}(\xi)|^2 |d\Phi|^2(\xi)$ for every finite interval $[A, B]$.

Proof. The Cramér reconstruction theorem for second-order stationary processes ([3], §7.5) asserts: given a non-negative finite Borel measure ν on a compact interval $I \subset \mathbb{R}$, there exists an orthogonal complex Borel measure $d\Phi$ on I with $|d\Phi|^2 = \nu$, and the process $u \mapsto \int_I e^{i\xi u} d\Phi(\xi)$ has spectral measure ν on I . Reality of Y on $\mathbb{R}$ is equivalent to the Hermitian symmetry $d\Phi(-\xi) = d\Phi(\xi)$, which is compatible with ν_∞ since $|K_T(-\mu)| = |K_T(\mu)|$ (from Y real) makes P_T even in μ and ν_∞ even on $[-1, 1]$. Applied to $\nu = \nu_\infty \cdot C$ (with the absolute constant $C = \lim_T \int_{\mathbb{R}} P_T d\mu = 2$ from (35) restoring physical normalization), the reconstruction gives (38). Uniqueness of $d\Phi$ given $|d\Phi|^2$ and Hermitian symmetry on $[-1, 1]$ follows from the standard Gram–Schmidt construction of orthogonal measures ([3], loc. cit.). $\square$

5 Entire Extension and Laguerre–Pólya Membership

Corollary 3 (Entire extension of exponential type ≤ 1). *Y extends to an entire function of exponential type ≤ 1 :*

$$Y(z) = \int_{-1}^1 e^{i\xi z} d\Phi(\xi), \quad |Y(z)| \leq \|d\Phi\| e^{|\operatorname{Im} z|}, \quad (39)$$

where $d\Phi$ is the Cramér spectral measure from (38) and $\|d\Phi\|$ is its total variation.

Proof. Define $Y(z) := \int_{-1}^1 e^{i\xi z} d\Phi(\xi)$ for $z \in \mathbb{C}$. The integrand is entire in z for each fixed $\xi \in [-1, 1]$; the measure $d\Phi$ is finite (total variation $\|d\Phi\| < \infty$) and the integrand is uniformly bounded on compact subsets of $\mathbb{C}$ by e^R for $|z| \leq R$ since $|\xi| \leq 1$. By the dominated convergence theorem and Morera's theorem, Y is entire. For the exponential-type bound,

$$|Y(z)| \leq \int_{-1}^1 |e^{i\xi z}| |d\Phi|(\xi) = \int_{-1}^1 e^{-\xi \operatorname{Im} z} |d\Phi|(\xi) \leq e^{|\operatorname{Im} z|} \int_{-1}^1 |d\Phi|(\xi) = \|d\Phi\| e^{|\operatorname{Im} z|}. \quad (40)$$

Restricted to $z = u \in \mathbb{R}$, (39) reproduces (38); uniqueness of the entire extension of the real-analytic function $u \mapsto Y(u)$ (with globally convergent representation) identifies this extension with Y on $\mathbb{R}$. $\square$

Lemma 6 (Non-negativity of the spectral measure). *The spectral measure $|d\Phi|^2$ of the representation (38) is a non-negative finite Borel measure on $[-1, 1]$.*

Proof. From (38), the Cramér covariance of Y is $C(\eta) = \int_{-1}^1 e^{i\xi\eta} d|\Phi|^2(\xi)$ with $|d\Phi|^2$ non-negative and supported in $[-1, 1]$ by construction. The power spectral density S from (14) equals the Radon–Nikodym derivative of the absolutely continuous part of $|d\Phi|^2$; Theorem 1 gives $S(\mu) = 0$ for $|\mu| > 1$, consistent with the support containment. $\square$

Theorem 2 (Akhiezer). *A real-valued entire function F of exponential type $\leq \tau$ whose spectral measure $|\hat{F}|^2$ is a non-negative measure supported in $[-\tau, \tau]$ belongs to the Laguerre–Pólya class. All zeros of F in $\mathbb{C}$ are real.*

Proof. [7], Lecture 16, Theorem 3; [4], §V.4. $\square$

Corollary 4 (Reality of zeros of Y). *All zeros of Y in $\mathbb{C}$ are real.*

Proof. Y is real on $\mathbb{R}$ (Definition 2), entire of exponential type ≤ 1 (Corollary 3), with spectral measure $|d\Phi|^2$ non-negative and supported in $[-1, 1]$ (Lemma 6). Apply Theorem 2. $\square$

6 Transfer to the Hardy Z-Function on the Real Line

The transfer uses only the real-line identity

$$Z(t) = Y(\Theta(t)) \sqrt{\Theta'(t)}, \quad t \in \mathbb{R}, \quad (41)$$

which is the rearrangement of (8) after substituting $u = \Theta(t)$. On $\mathbb{R}$, $\Theta'(t) > 0$ (Lemma 3) and $\sqrt{\Theta'(t)}$ is the principal (positive) square root of a strictly positive real number; no branch choice is involved.

Theorem 3 (Zero-set bijection under the unitary time change on $\mathbb{R}$). *Let c be admissible. The map $t \mapsto \Theta(t)$ is a bijection $\mathbb{R} \rightarrow \mathbb{R}$ that restricts to a bijection of zero sets,*

$$\{t \in \mathbb{R} : Z(t) = 0\} \longleftrightarrow \{u \in \mathbb{R} : Y(u) = 0\}, \quad u = \Theta(t), \quad (42)$$

with preservation of multiplicity: if $t_0 \in \mathbb{R}$ and Y has a zero of order m at $\Theta(t_0)$, then Z has a zero of order exactly m at t_0 , and conversely.

Proof. $\sqrt{\Theta'(t)}$ is strictly positive on $\mathbb{R}$. From (41), for any $t_0 \in \mathbb{R}$,

$$Z(t_0) = 0 \iff \sqrt{\Theta'(t_0)} Y(\Theta(t_0)) = 0 \iff Y(\Theta(t_0)) = 0. \quad (43)$$

Since $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection (Lemma 3), $t_0 \mapsto \Theta(t_0)$ is a bijection between the real zero set of Z and the real zero set of Y .

For multiplicity, let $t_0 \in \mathbb{R}$ and suppose Y has a zero of order m at $s_0 := \Theta(t_0)$. Since Y is entire (Corollary 3), there exists an entire g with $g(s_0) \neq 0$ and

$$Y(s) = (s - s_0)^m g(s). \quad (44)$$

Substitute $s = \Theta(t)$ (defined and C^∞ for $t \in \mathbb{R}$):

$$Y(\Theta(t)) = (\Theta(t) - \Theta(t_0))^m g(\Theta(t)). \quad (45)$$

The Hadamard-type identity

$$\Theta(t) - \Theta(t_0) = (t - t_0) h(t), \quad h(t) := \int_0^1 \Theta'(t_0 + u(t - t_0)) du, \quad (46)$$

valid for all $t \in \mathbb{R}$ because $\Theta \in C^1(\mathbb{R})$, gives $h(t_0) = \Theta'(t_0) > 0$, and continuity of h together with $\Theta' > 0$ on $\mathbb{R}$ ensures $h(t) > 0$ for t in a neighborhood of t_0 . Combining (41), (45), and (46),

$$Z(t) = Y(\Theta(t)) \sqrt{\Theta'(t)} = (t - t_0)^m h(t)^m g(\Theta(t)) \sqrt{\Theta'(t)}. \quad (47)$$

Denote the trailing factor

$$H(t) := h(t)^m g(\Theta(t)) \sqrt{\Theta'(t)}. \quad (48)$$

Then H is continuous at t_0 and

$$H(t_0) = \Theta'(t_0)^m g(s_0) \sqrt{\Theta'(t_0)} \neq 0 \quad (49)$$

because $\Theta'(t_0) > 0$ and $g(s_0) \neq 0$. Therefore Z has a zero of order exactly m at t_0 .

The converse (given a zero of Z of order m at $t_0 \in \mathbb{R}$, Y has a zero of order m at $\Theta(t_0)$) is obtained by applying the same argument with Θ replaced by Θ^{-1} , which satisfies $(\Theta^{-1})'(u) = 1/\Theta'(\Theta^{-1}(u)) > 0$ for $u \in \mathbb{R}$ by Lemma 3 and the inverse function theorem. $\square$

Corollary 5 (Zeros of Z on $\mathbb{R}$). *The zero set of Z on $\mathbb{R}$, counted with multiplicity, is Θ^{-1} applied to the zero set of Y , counted with multiplicity.*

Proof. Corollary 4 gives that every zero of the entire function Y lies on $\mathbb{R}$. Theorem 3 gives a multiplicity-preserving bijection between the zero set of Y on $\mathbb{R}$ and the zero set of Z on $\mathbb{R}$, implemented by $t \mapsto \Theta(t)$. $\square$

7 The Zero Locus of ζ on the Critical Line

Corollary 6 (Zeros of ζ on the critical line). *The zero set of ζ on the critical line $\{\operatorname{Re} s = \frac{1}{2}\}$, counted with multiplicity, is the image of $\Theta^{-1}(\{u \in \mathbb{R} : Y(u) = 0\})$ under the map $t \mapsto \frac{1}{2} + it$, with multiplicities inherited from those of Y .*

Proof. For $t \in \mathbb{R}$, $|e^{i\theta(t)}| = 1$, so $e^{i\theta(t)}$ is a nonzero complex number; hence $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)} Z(t)$ gives $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$, with matching orders of vanishing at every $t \in \mathbb{R}$ (multiplication by the non-vanishing real-analytic factor $e^{-i\theta(t)}$ does not alter the order of a zero of the real-analytic factor $Z(t)$ at a real point). Corollary 5 identifies the real zero set of Z with Θ^{-1} applied to the zero set of Y . The map $t \mapsto \frac{1}{2} + it$ is a bijection $\mathbb{R} \rightarrow \{\operatorname{Re} s = \frac{1}{2}\}$. Composing gives the stated identification. $\square$

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